

Edge Computing: Enhancing the IoT Experience

Edge computing pushes the frontiers of computing, data and services to the extremes of the network, where these are actually needed. It facilitates the collection and analysis of data at its source.



The Internet of Things has brought technology to every household. Consumers see a lot of value in connecting things, and to make sense of the collected data for enhancing their life style. IoT is also widely used in the industrial, automotive and commercial markets, as well as in emerging markets like 5G.

As per Gartner and other sources, the total number of connected devices worldwide today is around 20 billion. This number is expected to be 75 billion by 2025. With so many connected

sensors and devices, users and businesses must be able to analyse the data they produce at the right location and at the right time. However, most of the data management and analysis happens in the public cloud today, which has severe limitations with respect to latency, governance and security. The IoT industry has over the past few years adopted edge computing (and fog computing) to overcome these limitations, by placing the storage and computing nodes at the Internet's edge where the data source is.

Typical IoT architecture and limitations

Typical IoT architecture, as shown the Figure 1, processes the data from the sensors and devices in the public cloud. The IoT gateway, which is a dedicated physical device or a software program, is the connection point between the cloud and the intelligent devices. Its basic function is to pre-process data before sending it to the cloud. It can also autonomously control field devices based on data inputs by sensors. It connects the IoT devices to the Internet using IoT protocols.

